

Numerical Solution for 1D and 2D Diffusion Equation

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1D Diffusion: Mathematical Formulation

Problem statement (solving the 1D diffusion equation)

BOUNDARY CONDITIONS (BCs)

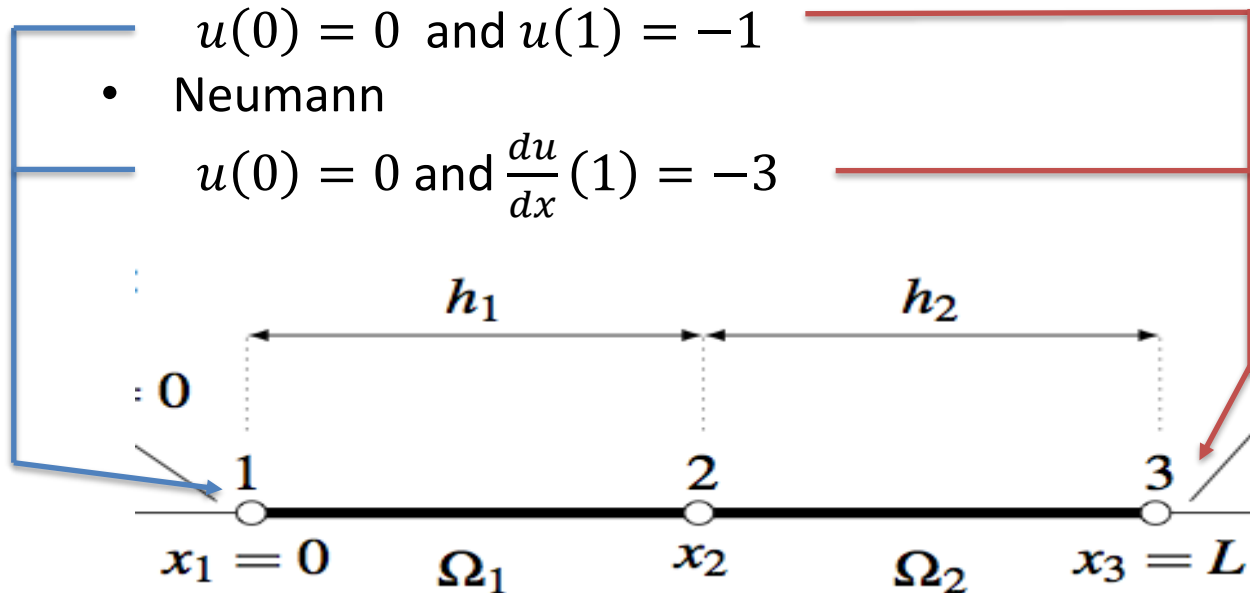
Two types :

- Dirichlet

$$u(0) = 0 \text{ and } u(1) = -1$$

- Neumann

$$u(0) = 0 \text{ and } \frac{du}{dx}(1) = -3$$



Given Equation :

$$\frac{d^2u}{dx^2} - 2u = f(x), 0 < x < 1$$

Where,

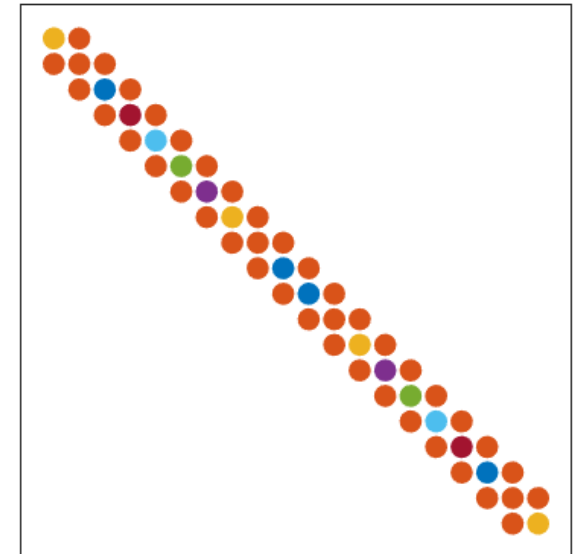
$$f(x) = 4x^2 - 2x - 4$$

The exact solution was :

$$u(x) = -2x^2 + x$$

Solved By:

1. FDM
2. FEM
3. FVM



1D Diffusion: Numerical Method

FDM Discretization

For Dirichlet-Dirichlet Boundary Conditions

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} - 2u_i = f_i$$



$$u_0 - (2 + 2h^2)u_1 + u_2 = h^2 f(x_1)$$

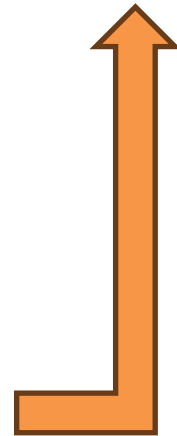
$$u_1 - (2 + 2h^2)u_2 + u_3 = h^2 f(x_2)$$



Rearranging them to be

$$-(2 + 2h^2)u_1 + u_2 = h^2 f(x_1) - u_0$$

$$u_1 - (2 + 2h^2)u_2 = h^2 f(x_2) - u_3$$



$$\begin{bmatrix} -(2 + 2h^2) & 1 \\ 1 & -(2 + 2h^2) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} h^2 f(x_1) - u_0 \\ h^2 f(x_2) - u_3 \end{Bmatrix}$$



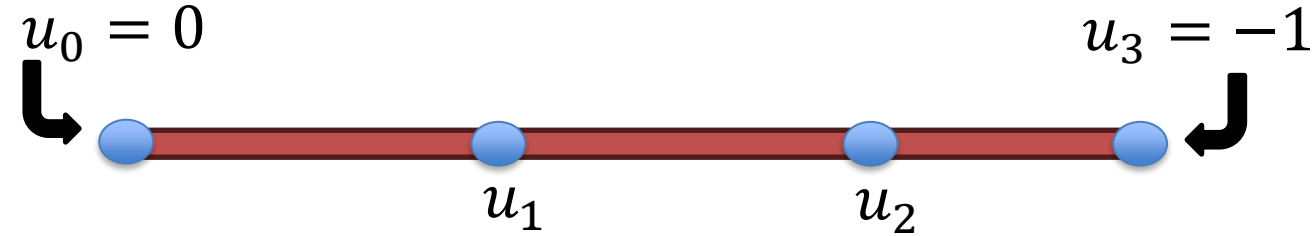
Exact Solution

$$u(x) = -2x^2 + x$$

u_0	0
u_1	1/9
u_2	-2/9
u_3	-1

FDM Solution

u_0	0
u_1	1/9
u_2	-2/9
u_3	-1



1D Diffusion: Numerical Method

FDM Discretization

For Dirichlet-Neumann Boundary Conditions

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} - 2u_i = f_i$$

$$u_0 - (2 + 2h^2)u_1 + u_2 = h^2 f(x_1)$$

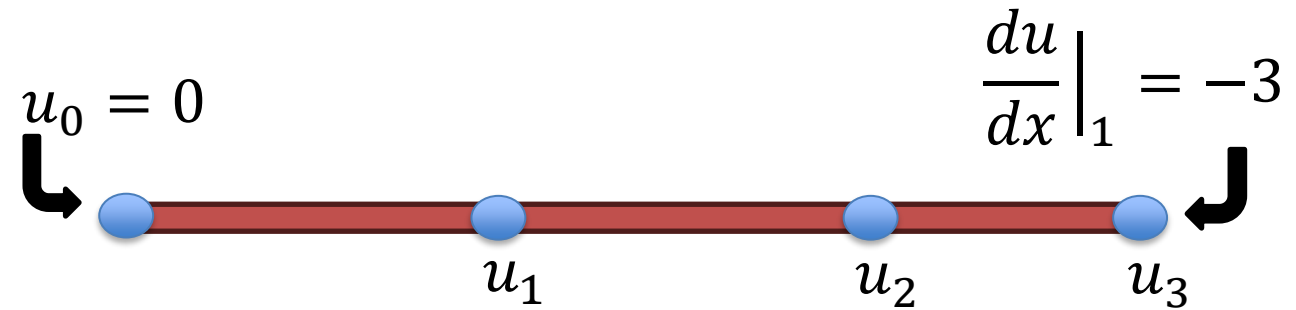
$$u_1 - (2 + 2h^2)u_2 + u_3 = h^2 f(x_2)$$

$$\frac{u_4 - u_2}{2h} = \frac{du}{dx} = -3$$

$$u_4 = u_2 - 6h$$

$$u_4 - (2 + 2h^2)u_3 + u_2 = h^2 f(x_3)$$

$$(2u_2 - 6h) - (2 + 2h^2)u_3 = h^2 f(x_3)$$



$$\begin{bmatrix} -(2 + 2h^2) & 1 & 0 \\ 1 & -(2 + 2h^2) & 1 \\ 0 & 2 & -(2 + 2h^2) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} h^2 f(x_1) - u_0 \\ h^2 f(x_2) \\ h^2 f(x_3) + 6h \end{Bmatrix}$$

Exact Solution

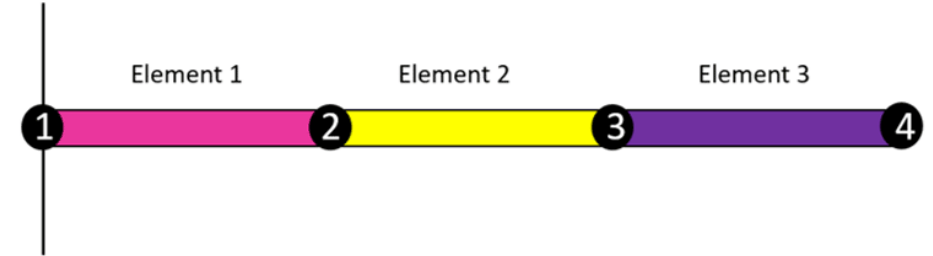
$$u(x) = -2x^2 + x$$

u_0	0
u_1	1/9
u_2	-2/9
u_3	-1

FDM Solution

u_0	0
u_1	1/9
u_2	-2/9
u_3	-1

1D Diffusion: Numerical Method



FEM Methods

Using the shape functions

$$u^{(e)}(x) = N_i(x)u_i + N_j(x)u_j$$

$$N_i(x) = \frac{x_j - x}{h}$$

$$N_j(x) = \frac{x - x_i}{h}$$

Galerkin Method

$$\int_0^1 W \cdot R(x) dx = 0$$

$$\int_0^1 w \left(\frac{d^2 u}{dx^2} - 2u - f(x) \right) dx = 0$$

- Weak form

$$\int_0^1 \left(\frac{dW}{dx} \frac{du}{dx} + 2Wu \right) dx = - \int_0^1 W f(x) dx + \left[W \frac{du}{dx} \right]_0^1$$

- Stiffness matrix

$$[k^e] = \int_e \left(\begin{bmatrix} \frac{dN_i}{dx} \\ \frac{dN_j}{dx} \end{bmatrix} \begin{bmatrix} \frac{dN_i}{dx} & \frac{dN_j}{dx} \end{bmatrix} + 2 \begin{bmatrix} N_i \\ N_j \end{bmatrix} \begin{bmatrix} N_i & N_j \end{bmatrix} \right) dx$$

$$[k^e] = [k_1^e] + [k_2^e]$$

- Element Force Vector

$$\{f^e\} = - \int_e W^T f(x) dx = - \int_{x_i}^{x_j} \begin{Bmatrix} N_i(x) \\ N_j(x) \end{Bmatrix} (4x^2 - 2x - 4) dx$$

1D Diffusion: Numerical Method

Global Stiffness Matrix

$$[K] = \begin{bmatrix} \frac{1}{h} + \frac{2h}{3} & -\frac{1}{h} + \frac{2h}{6} & 0 & 0 \\ -\frac{1}{h} + \frac{2h}{6} & \left(\frac{1}{h} + \frac{2h}{3}\right) + \left(\frac{1}{h} + \frac{2h}{3}\right) & -\frac{1}{h} + \frac{2h}{6} & 0 \\ 0 & -\frac{1}{h} + \frac{2h}{6} & \left(\frac{1}{h} + \frac{2h}{3}\right) + \left(\frac{1}{h} + \frac{2h}{3}\right) & -\frac{1}{h} + \frac{2h}{6} \\ 0 & 0 & -\frac{1}{h} + \frac{2h}{6} & \frac{1}{h} + \frac{2h}{3} \end{bmatrix}$$

Force Vectors

$$\{F\} = \begin{Bmatrix} f_1^1 \\ f_2^1 + f_1^2 \\ f_2^2 + f_1^3 \\ f_2^3 \end{Bmatrix}$$

Solving $KU = F + G$

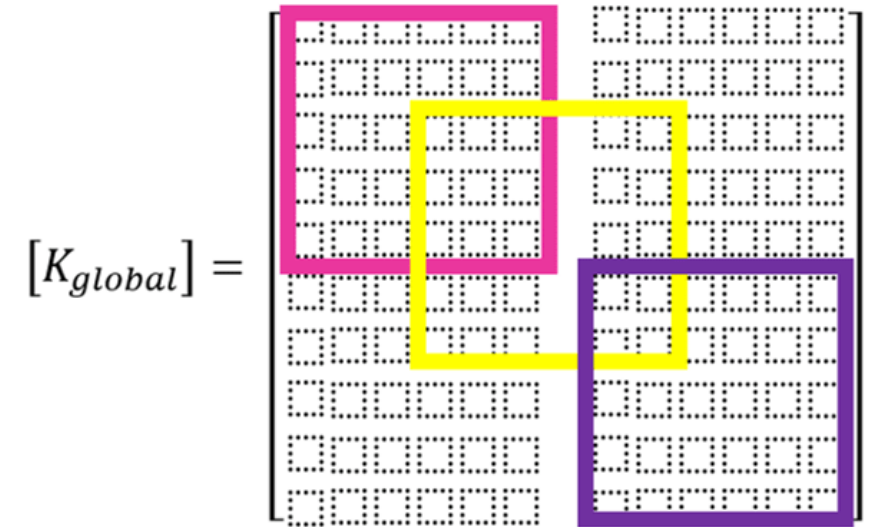
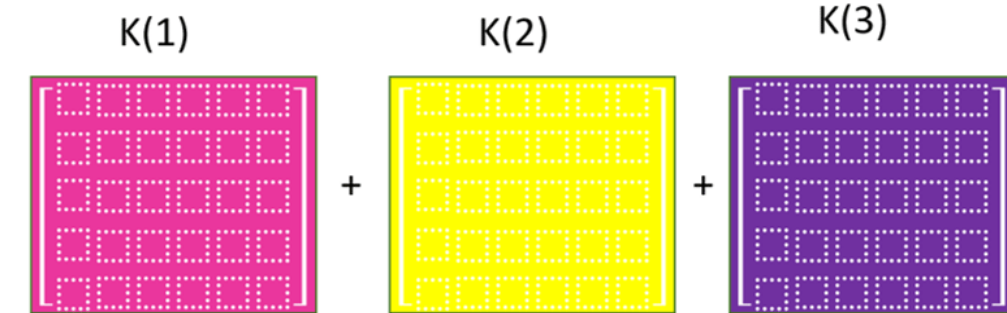
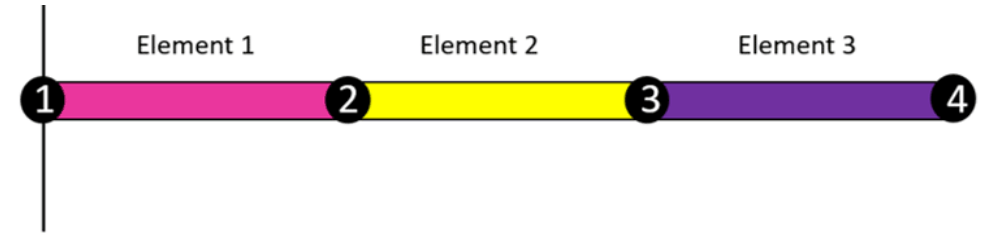
$$G = \begin{bmatrix} G_1^1 \\ G_2^1 + G_1^2 \\ G_2^2 + G_1^3 \\ G_2^3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3 \end{bmatrix}$$

For Dirichlet-Dirichlet BCs

$$U = \begin{bmatrix} 0 \\ 0.1180556 \\ -0.2152778 \\ -1 \end{bmatrix}$$

For Dirichlet-Neumann BCs

$$U = \begin{bmatrix} 0 \\ 0.12314452 \\ -0.20392548 \\ -0.97976461 \end{bmatrix}$$



1D Diffusion: Numerical Method

FVM Methods

$$\int_{i-\frac{1}{2}}^{i+\frac{1}{2}} \frac{d^2 u}{dx^2} dx - \int_{i-\frac{1}{2}}^{i+\frac{1}{2}} 2u dx = \int_{i-\frac{1}{2}}^{i+\frac{1}{2}} f(x) dx$$

$$\left[\frac{du}{dx} \right]_{i+\frac{1}{2}} - \left[\frac{du}{dx} \right]_{i-\frac{1}{2}} - \int_{i-\frac{1}{2}}^{i+\frac{1}{2}} 2u dx = \int_{i-\frac{1}{2}}^{i+\frac{1}{2}} f(x) dx$$

$$\left(\frac{u_{i+1} - u_i}{h} - \frac{u_i - u_{i-1}}{h} \right) - 2u_i h = f(x_i)h$$

For node 1

$$\frac{u_2}{h} - \left(\frac{2}{h} + 2h \right) u_1 + \frac{u_0}{h} = f(x_1)h$$

$$\frac{u_2}{h} - \left(\frac{2}{h} + 2h \right) u_1 = hf(x_1) - \frac{u_0}{h}$$

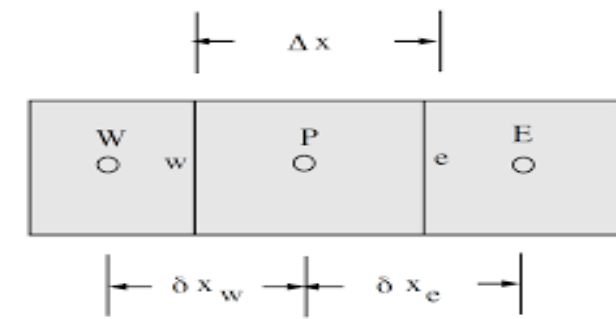


Figure 2.11: Arrangement of Control Volumes



For Dirichlet-Dirichlet BCs

At node 2

$$-\left(\frac{2}{h} + 2h \right) u_2 + \frac{u_1}{h} = hf(x_2) - \frac{u_3}{h}$$

For Dirichlet-Neumann BCs

At node 3

$$-2(1 + h^2)U_3 + 2U_2 = f(x_3)h^2 + 6h$$

1D Diffusion: Numerical Method

For Dirichlet-Dirichlet Boundary Conditions

The stiffness matrix

$$A = \begin{bmatrix} -\left(\frac{2}{h} + 2h\right) & \frac{1}{h} \\ \frac{1}{h} & -\left(\frac{2}{h} + 2h\right) \end{bmatrix}$$

The right hand side of the matrix is

$$f = \begin{Bmatrix} hf(x_1) - \left(\frac{u_0}{h}\right) \\ hf(x_2) - \left(\frac{u_3}{h}\right) \end{Bmatrix}$$

Solving them we get the values of the interior nodes u as

$$u = \begin{bmatrix} 0 \\ 0.1111 \\ -0.2222 \\ -1 \end{bmatrix}$$

For Dirichlet-Neumann Boundary Conditions

The stiffness matrix

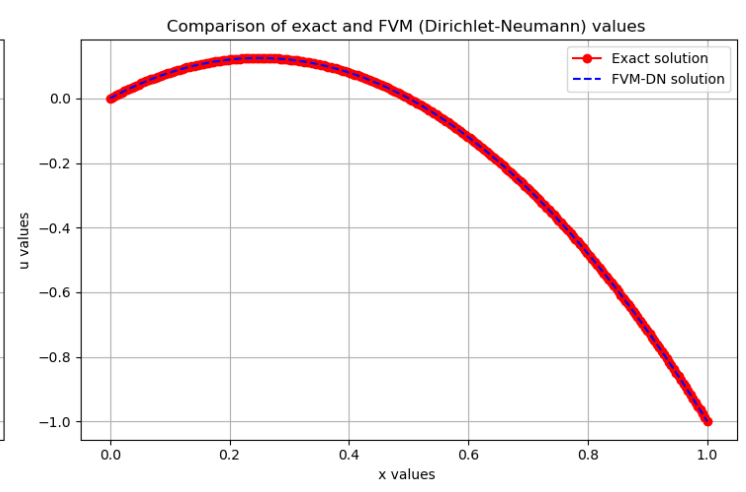
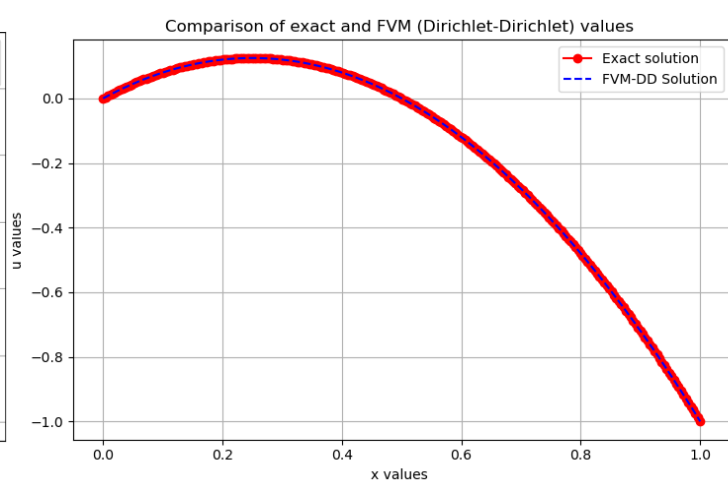
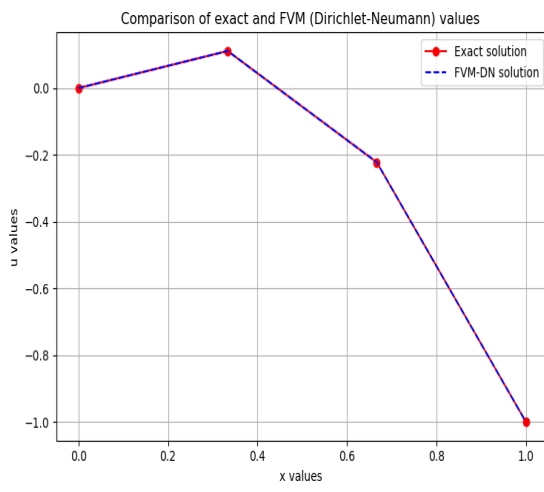
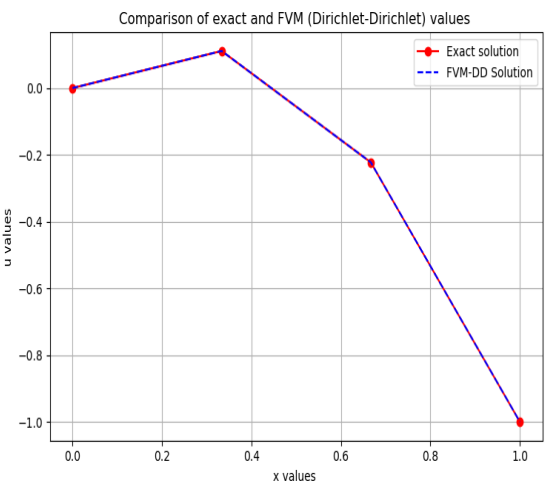
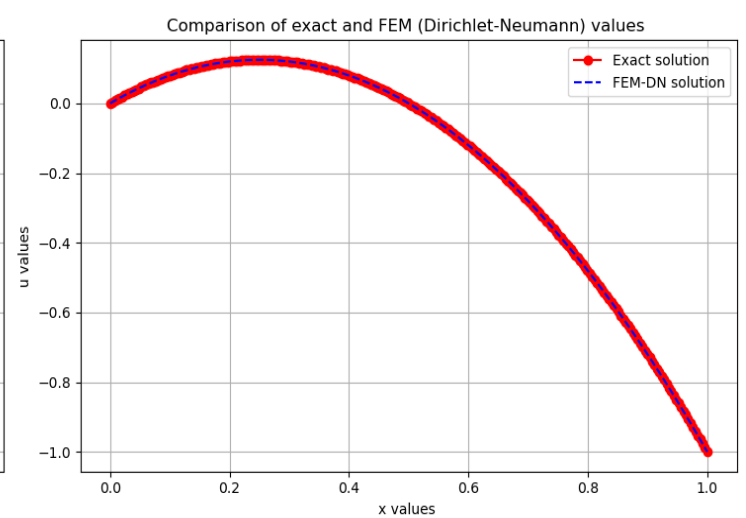
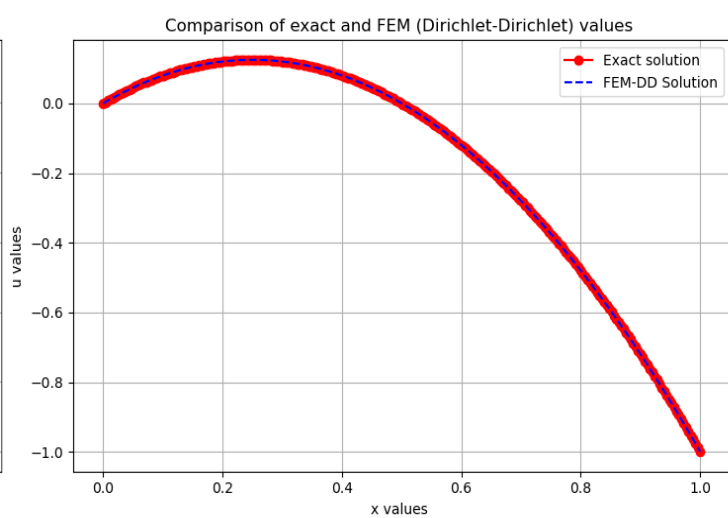
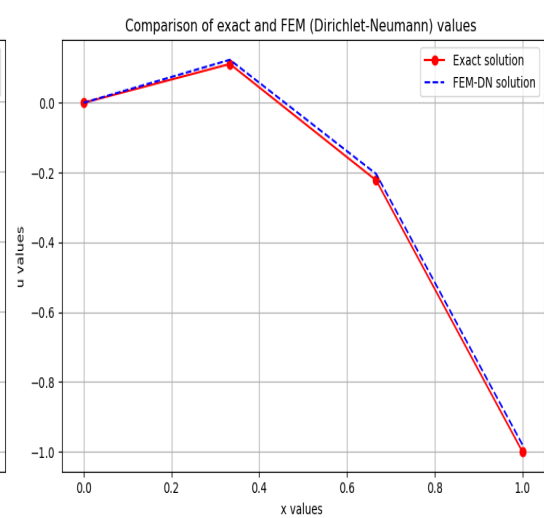
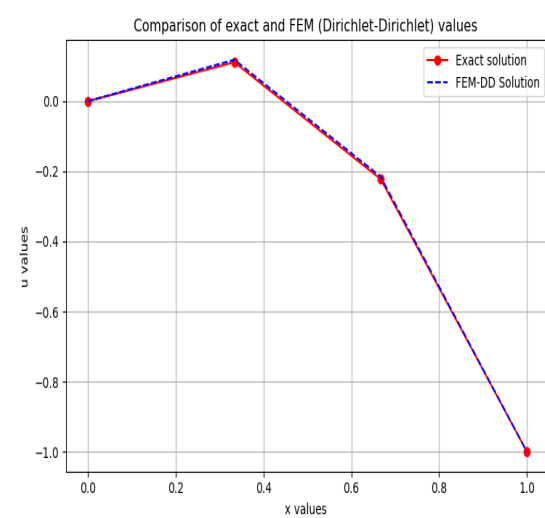
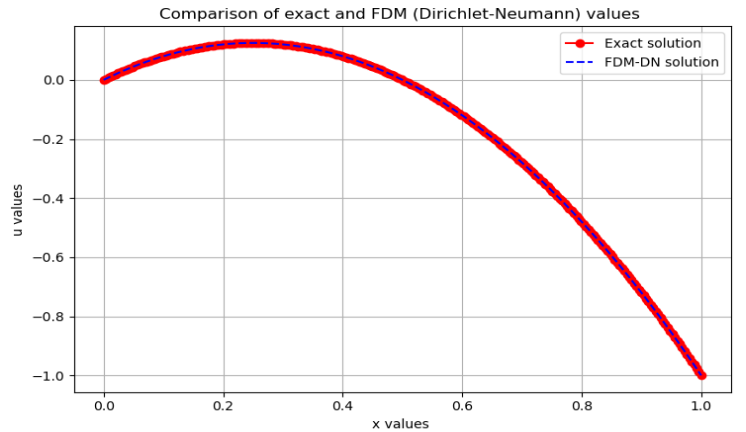
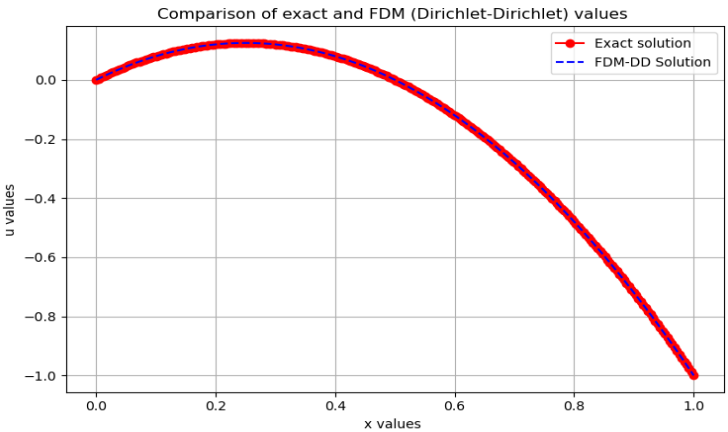
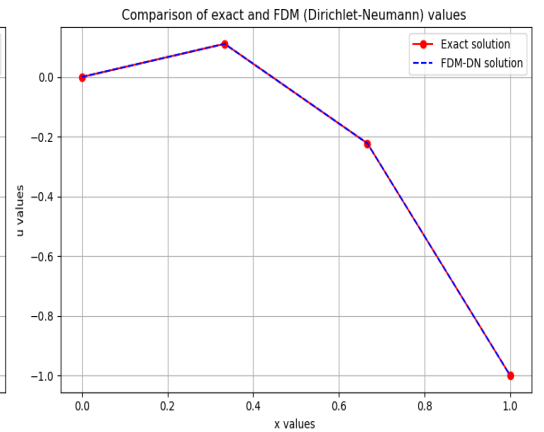
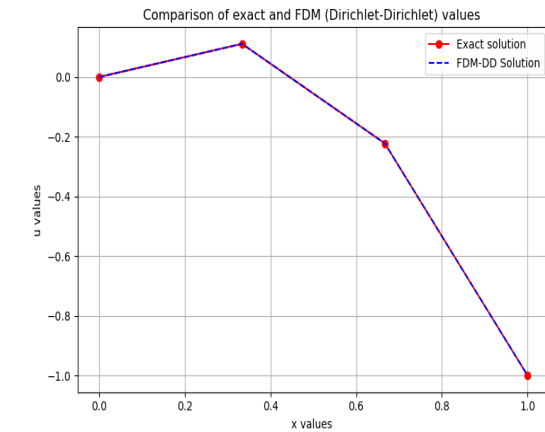
$$A = \begin{bmatrix} -(2 + 2h^2) & 1 & 0 \\ 1 & -(2 + 2h^2) & 1 \\ 0 & 2 & -2(1 + h^2) \end{bmatrix}$$

The right hand side of the matrix is

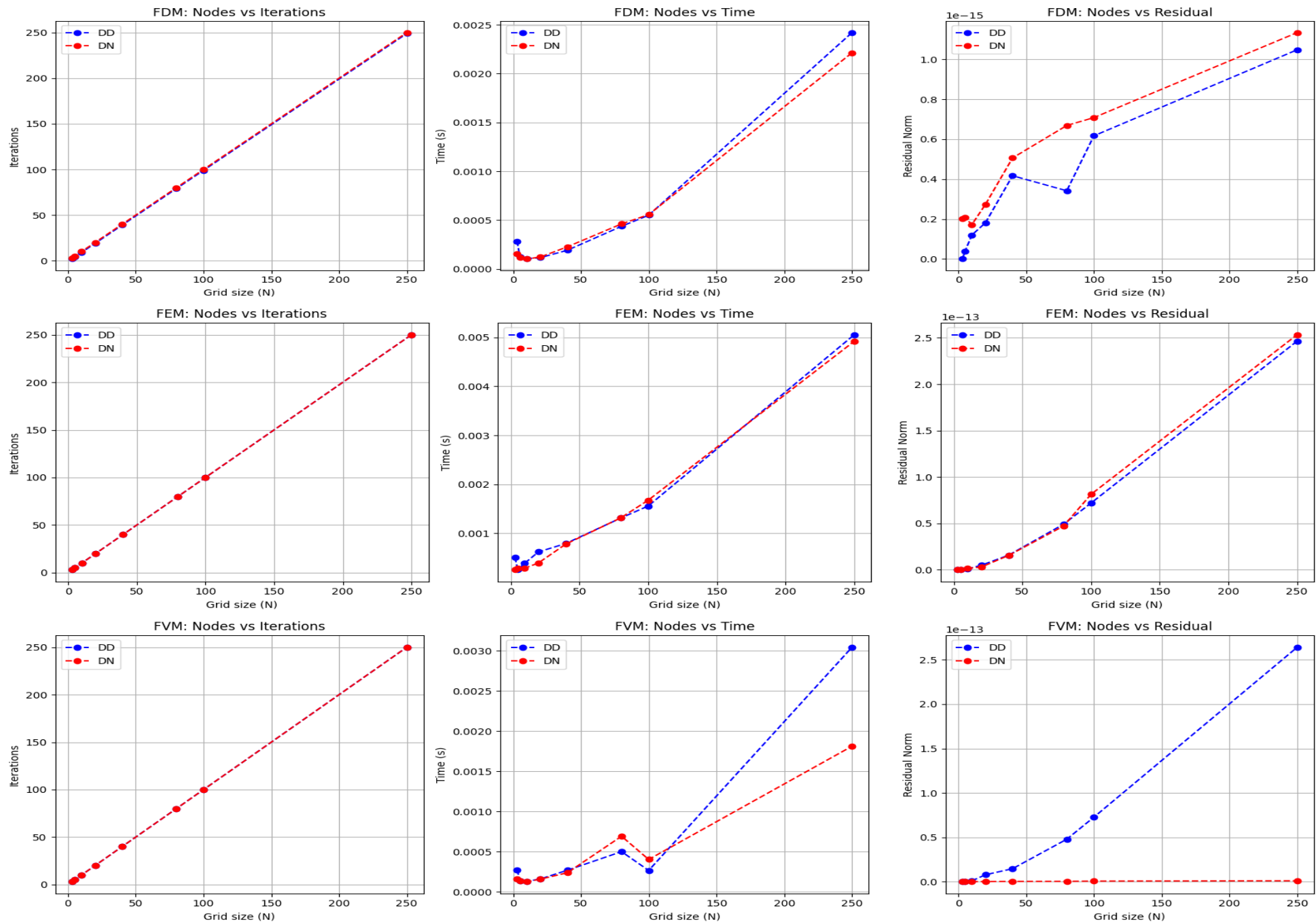
$$f = \begin{Bmatrix} h^2 f(x_1) - u_0 \\ h^2 f(x_2) \\ h^2 f(x_3) - 6h \end{Bmatrix}$$

Solving them we get the values of the interior nodes u as

$$u = \begin{bmatrix} 0 \\ 0.1111 \\ -0.2222 \\ -1 \end{bmatrix}$$



1D Diffusion: Results & Discussion



2D Diffusion: Mathematical Formulation

Solving the 2D-diffusion equation

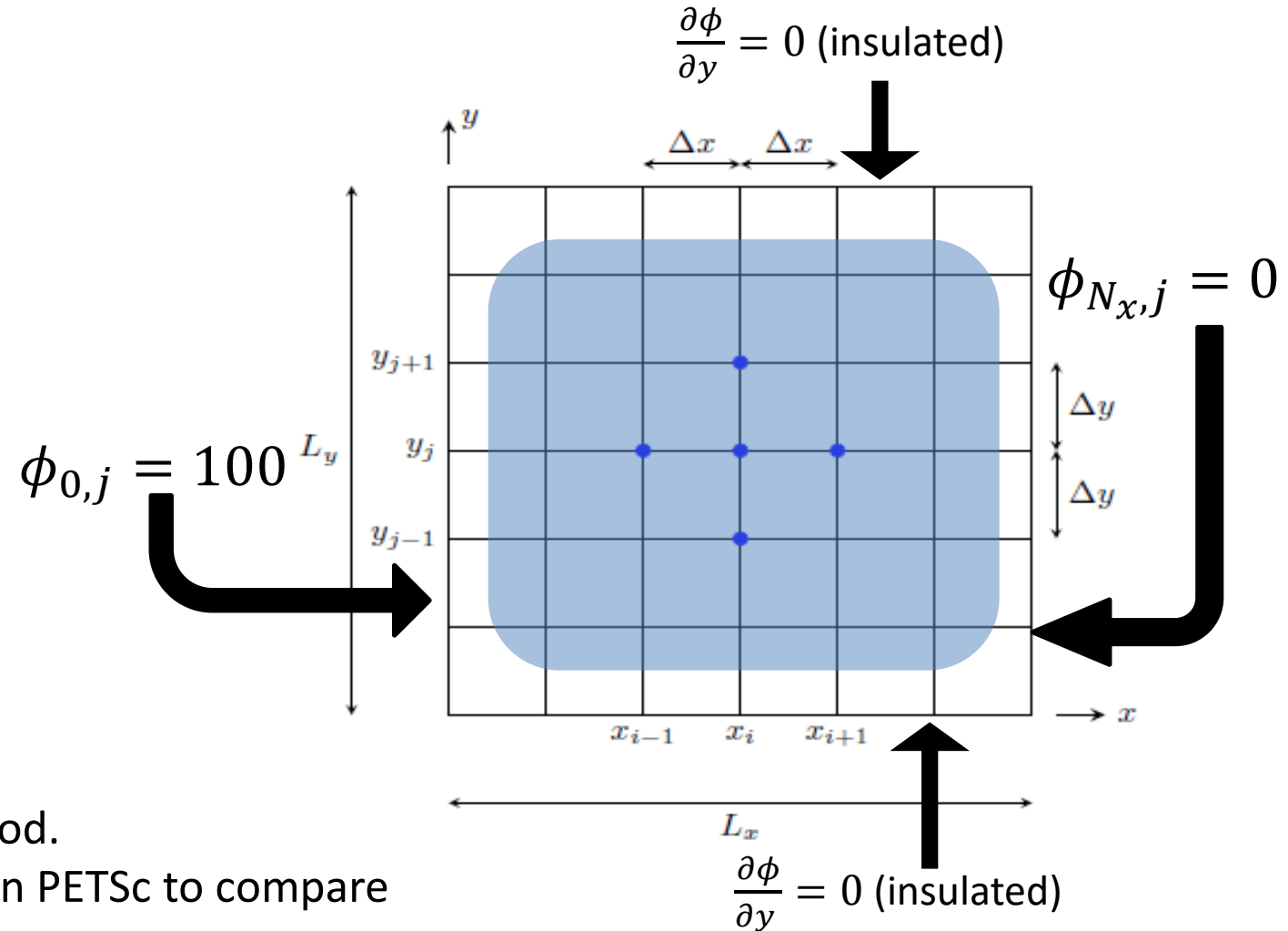
$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

The Given Boundary Conditions are :

- $\phi = 100$ at $x = 0$
- $\phi = 0$ at $x = L_x$
- Insulated boundaries $\left(\frac{\partial \phi}{\partial y} = 0\right)$ at $y = 0$ and $y = L_y$

Objectives:

1. To formulate using the Finite Difference Method.
2. To implement the iterative numerical solvers in PETSc to compare performance .
3. To analyse the effects of different grid resolutions on numerical solutions
4. TO visualize the steady-state results using contours and surface plots



2D Diffusion: Numerical Method

FDM Discretised equation

$$\frac{d^2 u}{dx^2} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2}$$

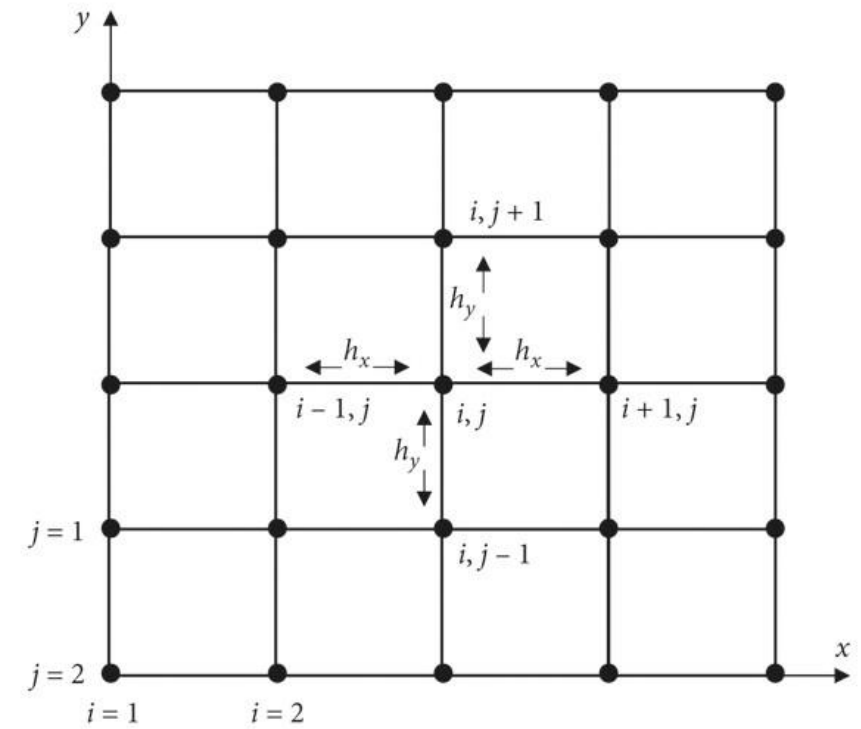
$$\frac{d^2 u}{dy^2} = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2}$$

Combining them leads to 5 point stencils of this expression

$$\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} = 0$$



$$u_{i+1,j} + u_{i-1,j} + \beta^2(u_{i,j+1} + u_{i,j-1}) - 2(1 + \beta^2)u_{i,j} = 0$$



For 5X5 grid points , we have :

- External grid or nodes are known
- Unknown internal grids
 - 16 known grid points
 - 9 unknown grid points
 - 9 equations to be formed

2D Diffusion: Numerical Method

Which leads to the formation of matrix

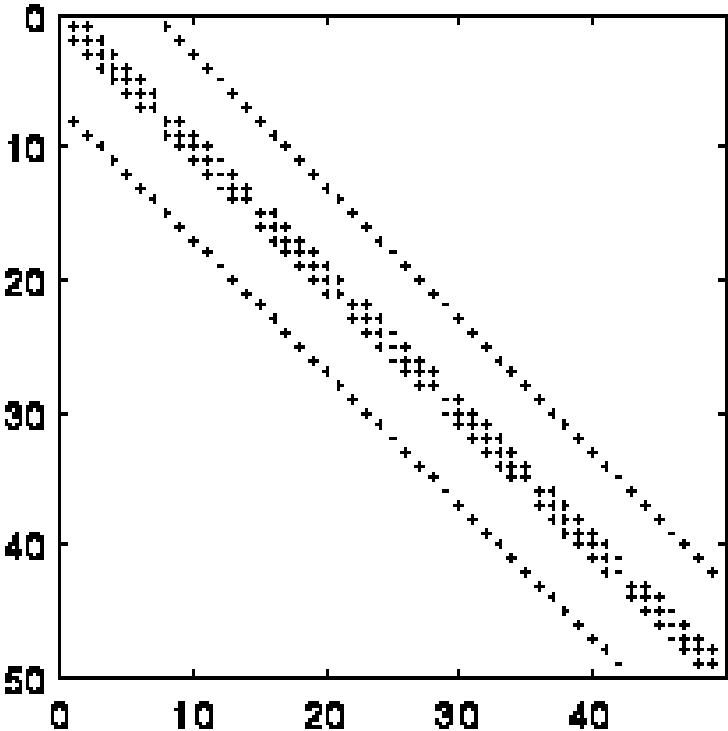
$$\begin{bmatrix} \gamma & 1 & 0 & 2\beta^2 & 0 & 0 & 0 & 0 & 0 \\ 1 & \gamma & 1 & 0 & 2\beta^2 & 0 & 0 & 0 & 0 \\ 0 & 1 & \gamma & 0 & 0 & 2\beta^2 & 0 & 0 & 0 \\ \beta^2 & 0 & 0 & \gamma & 1 & 0 & \beta^2 & 0 & 0 \\ 0 & \beta^2 & 0 & 1 & \gamma & 1 & 0 & \beta^2 & 0 \\ 0 & 0 & \beta^2 & 0 & 1 & \gamma & 0 & 0 & \beta^2 \\ 0 & 0 & 0 & 2\beta^2 & 0 & 0 & \gamma & 1 & 0 \\ 0 & 0 & 0 & 0 & 2\beta^2 & 0 & 1 & \gamma & 1 \\ 0 & 0 & 0 & 0 & 0 & 2\beta^2 & 0 & 1 & \gamma \end{bmatrix} \begin{bmatrix} u_{11} \\ u_{21} \\ u_{31} \\ u_{12} \\ u_{22} \\ u_{32} \\ u_{13} \\ u_{23} \\ u_{33} \end{bmatrix} = \begin{bmatrix} -\beta^2 u_{1,0} - u_{0,1} \\ -\beta^2 u_{2,0} \\ -\beta^2 u_{3,0} - u_{4,1} \\ -u_{0,2} \\ 0 \\ -u_{4,2} \\ -\beta^2 u_{1,4} - u_{0,3} \\ -\beta^2 u_{2,4} \\ -\beta^2 u_{3,4} - u_{4,3} \end{bmatrix}$$

$$\left[\begin{array}{c} \diagup \\ \diagdown \end{array} \right] A_i \left[\begin{array}{c} \diagup \\ \diagdown \end{array} \right] x_i = \left[\begin{array}{c} \diagup \\ \diagdown \end{array} \right] b_i$$

Where

And $\gamma = -2(1 + \beta^2)$

$$\beta = \frac{\Delta x}{\Delta y}$$



PETSc Workflow

Bash script

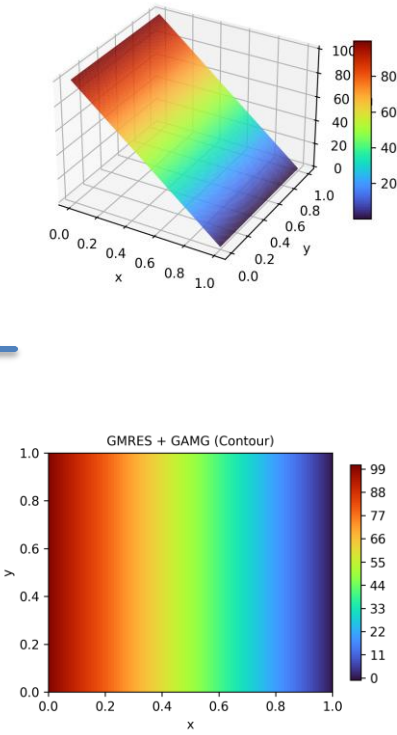
```
mpirun -np 20 ./petsc  
  
Nx = [5, 10, 20, 40, 80,  
100, 250 ]  
Nx = Ny
```

`petsc.cpp`

- 1. Setup Grid
- 2. Assemble Matrix (A)
- 3. Assemble Vectors (b)
- 4. Set KSP solver and Preconditioner
- 5. Solve and save file

- Perfromance.log
- Result_5X5.txt
- ...
- Result_250X250.txt

Visualization

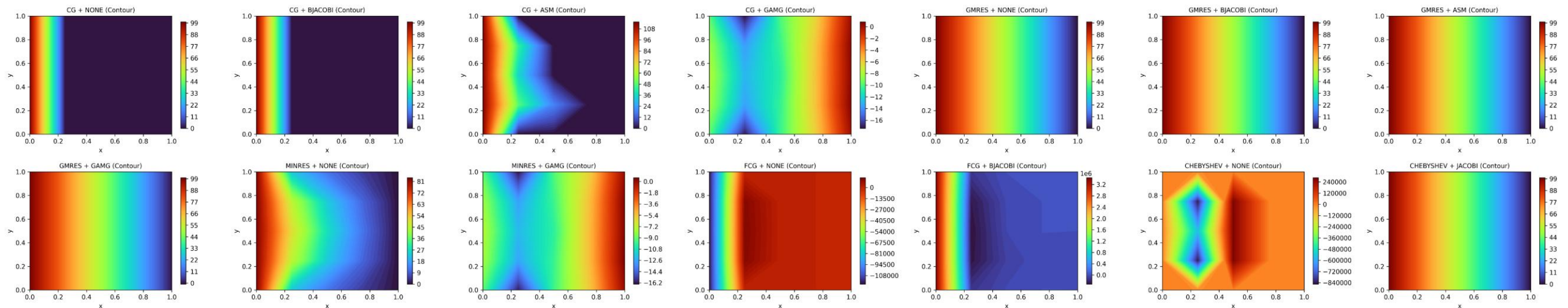


Solvers and Preconditioners used

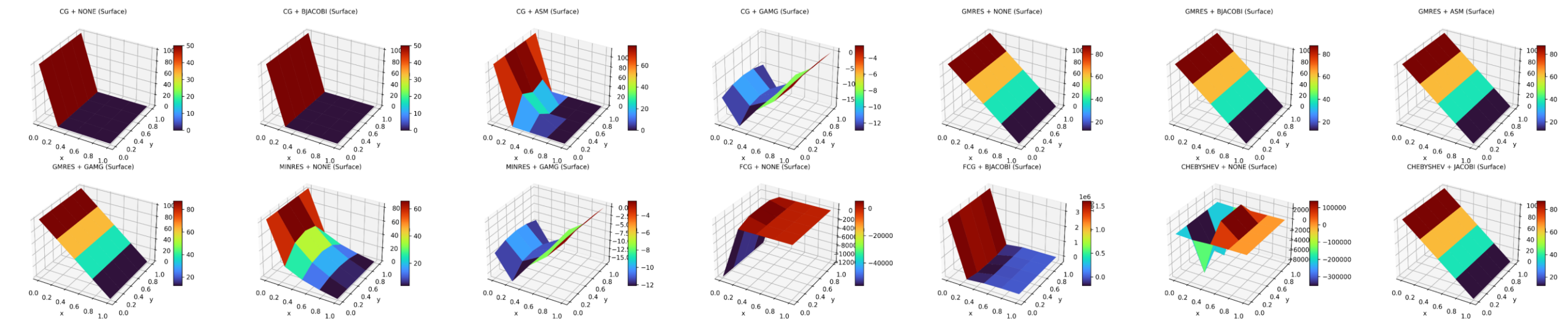
Solver	Preconditioner
CG	None
	Bjacobi
	ASM
	GAMG
GMRES	None
	Bjacobi
	ASM
	GAMG
MINRES	None
	GAMG
FCG	None
	Bjacobi
Chebyshev	None
	Jacobi

2D Diffusion: Results & Discussion

Contour Plots for Grid 5x5

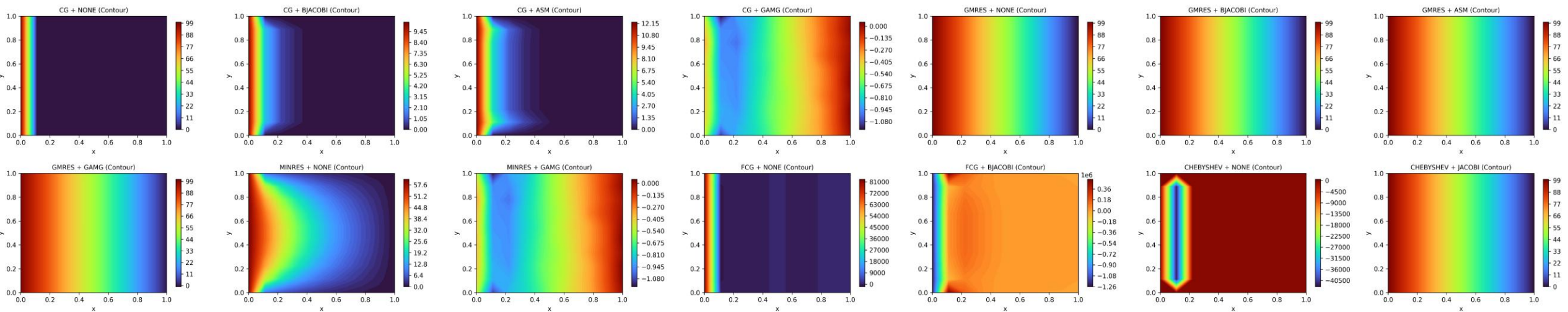


Surface Plots for Grid 5x5

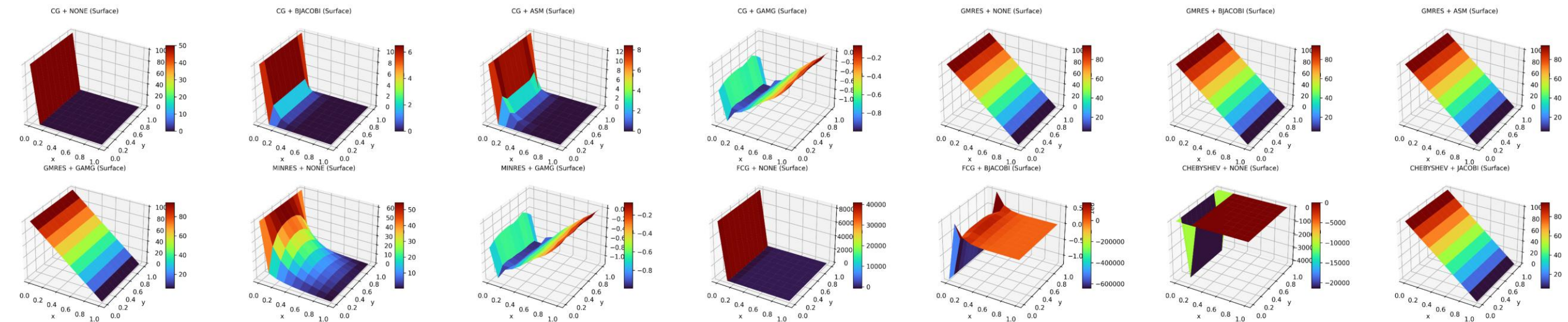


2D Diffusion: Results & Discussion

Contour Plots for Grid 10x10

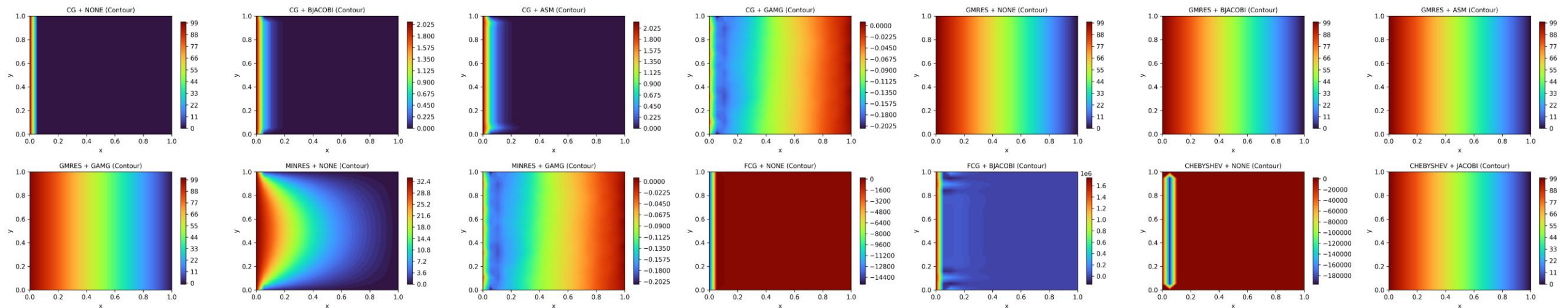


Surface Plots for Grid 10x10

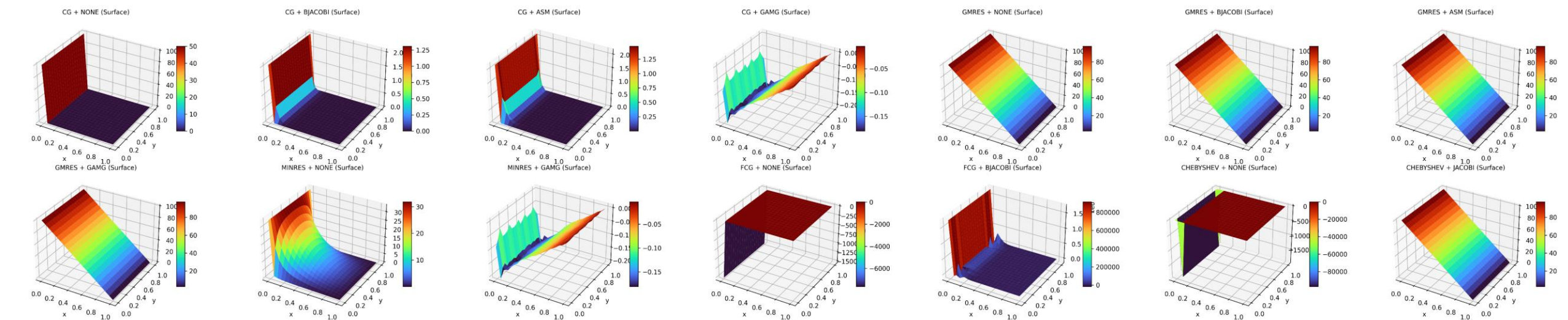


2D Diffusion: Results & Discussion

Contour Plots for Grid 20x20

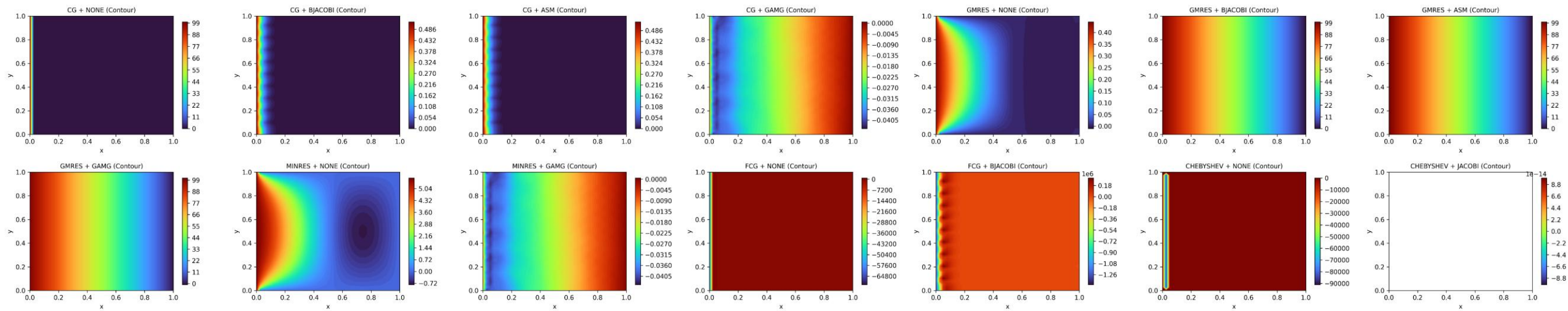


Surface Plots for Grid 20x20

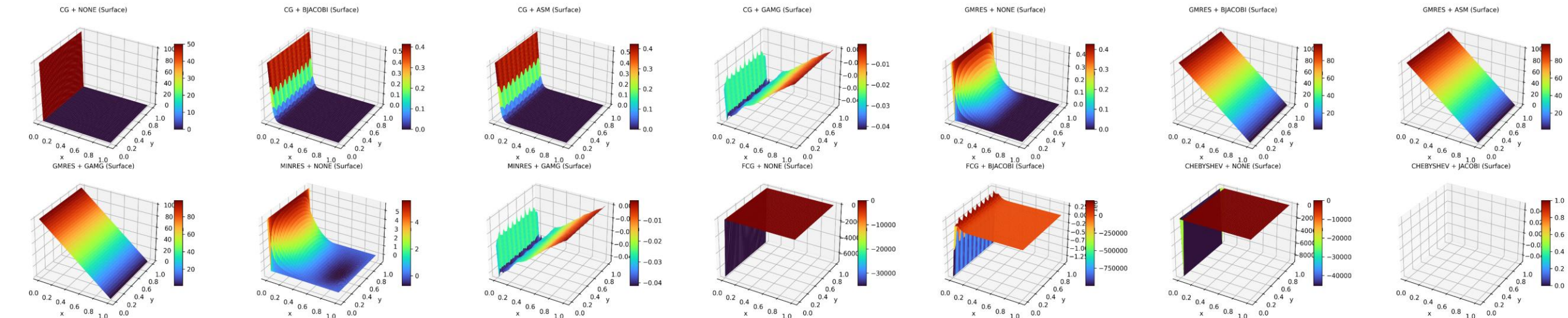


2D Diffusion: Results & Discussion

Contour Plots for Grid 40x40

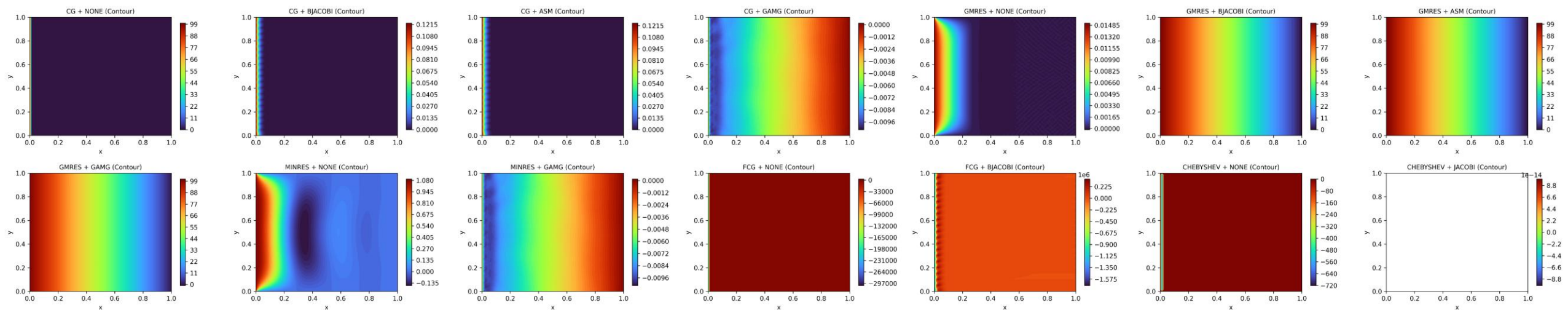


Surface Plots for Grid 40x40

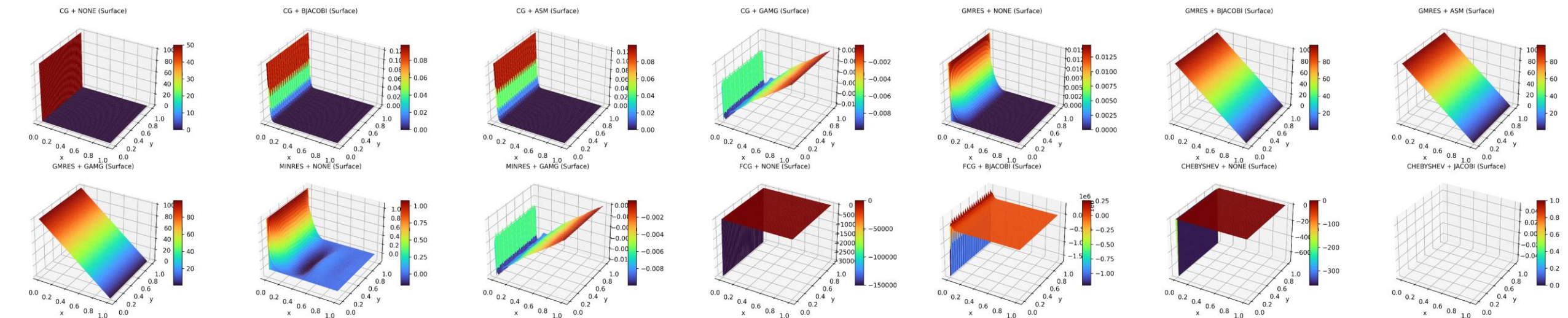


2D Diffusion: Results & Discussion

Contour Plots for Grid 80x80

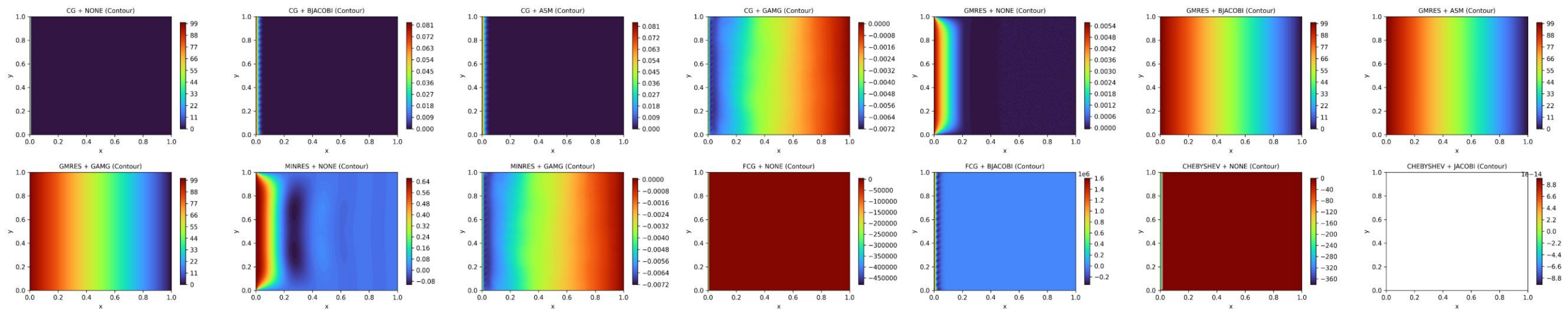


Surface Plots for Grid 80x80

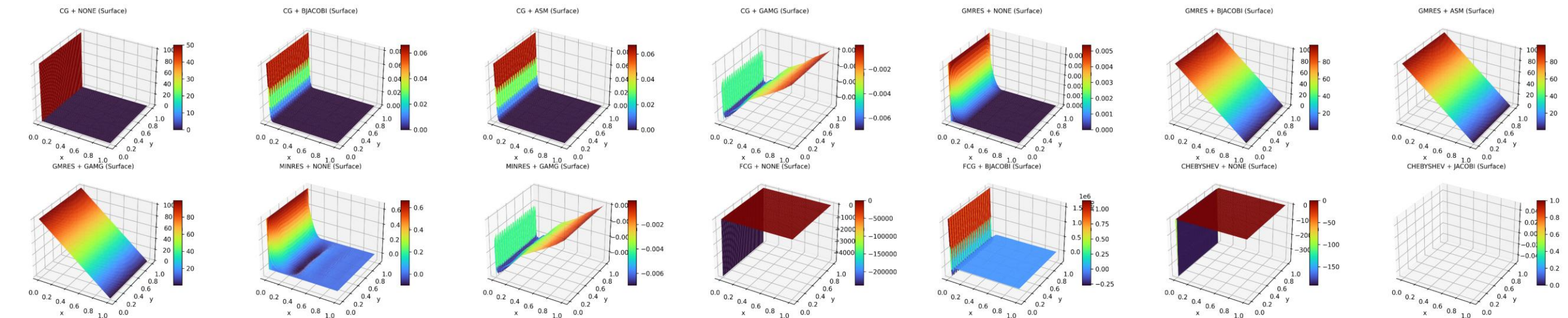


2D Diffusion: Results & Discussion

Contour Plots for Grid 100x100

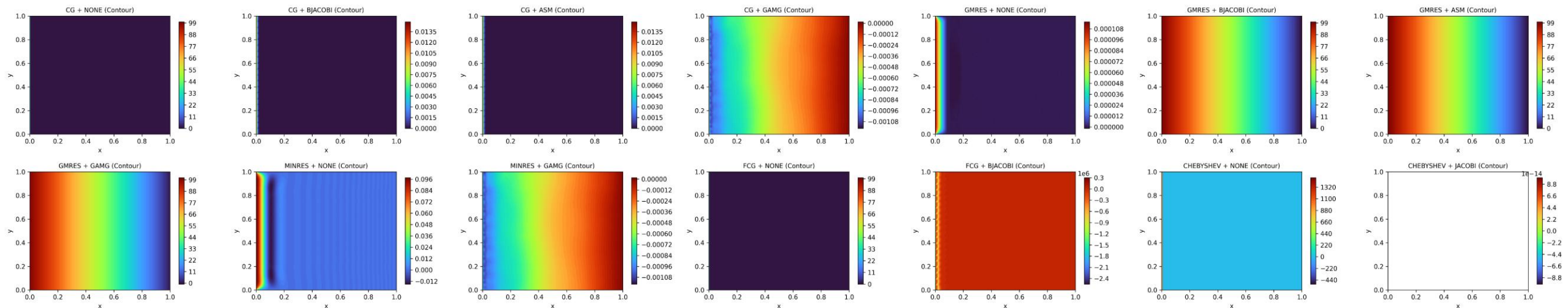


Surface Plots for Grid 100x100

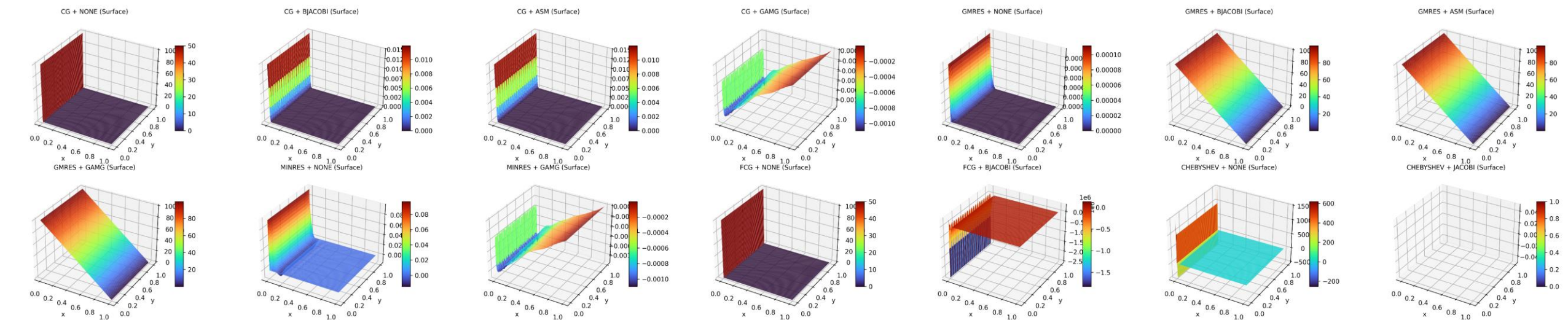


2D Diffusion: Results & Discussion

Contour Plots for Grid 250x250



Surface Plots for Grid 250x250

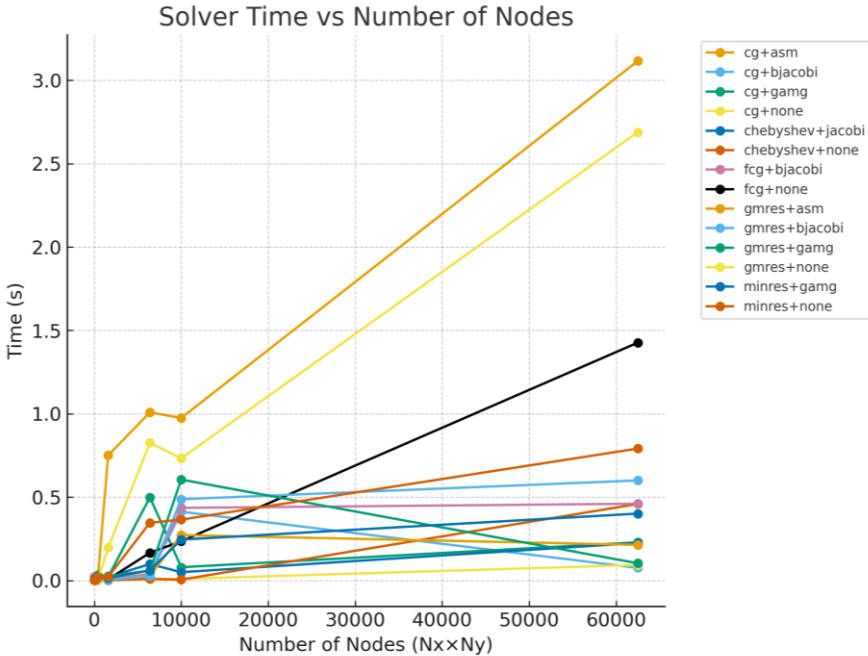
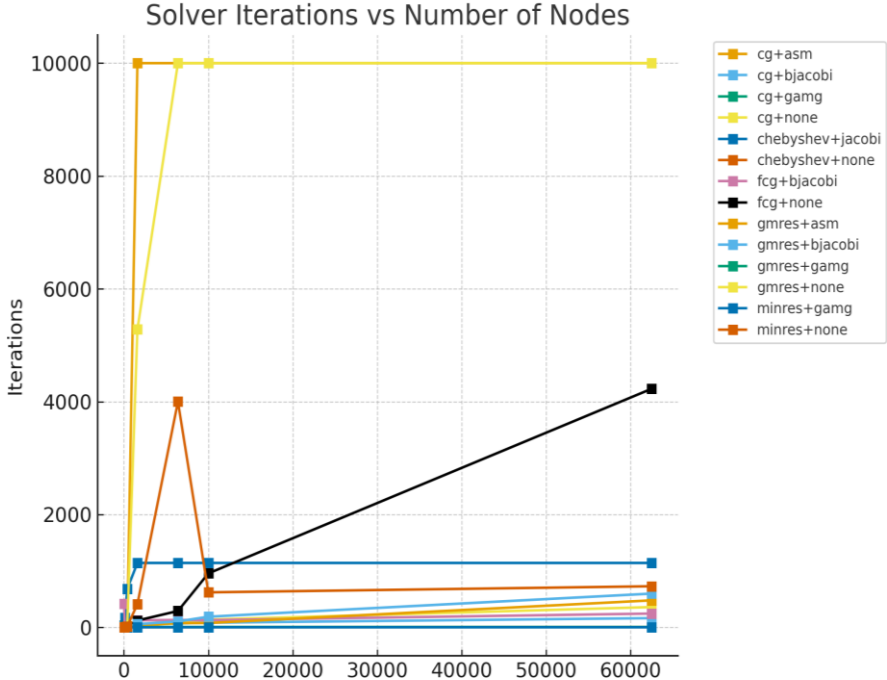


Performance Analysis

Nx	solver	Preconditioner	iterations	Time(S)	Residuals	Nx	Solver	Preconditioner	Iteration	Time(s)	Residuals	Nx	Solver	Preconditioner	Iteration	Time(s)	Residuals
5X5	cg	none	2	0.002892	2771.28	20X20	cg	none	2	0.002778	153159	80X80	cg	none	2	0.00658	5.51E+06
	cg	bjacobi	2	0.153307	44.7834		cg	bjacobi	2	0.013858	453.344		cg	bjacobi	2	0.236404	1004.21
	cg	asm	2	0.032191	95.1272		cg	asm	2	0.002112	461.613		cg	asm	2	0.180577	1016.7
	cg	gamg	2	0.010793	416.696		cg	gamg	2	0.015994	1767.29		cg	gamg	2	0.136916	7456.87
	gmres	none	10	0.000594	1.48E-09		gmres	none	1202	0.18796	0.004374		gmres	none	10000	0.688821	894.362
	gmres	bjacobi	14	0.001395	0.000876		gmres	bjacobi	36	0.013082	0.002933		gmres	bjacobi	109	0.114216	0.010018
	gmres	asm	9	0.001626	0.000456		gmres	asm	26	0.115159	0.001932		gmres	asm	74	0.061259	0.009642
	gmres	gamg	6	0.007711	0.001396		gmres	gamg	7	0.026423	0.005061		gmres	gamg	6	0.143268	0.039339
	minres	none	10000	0.292877	0.764782		minres	none	10000	0.423593	6.98421		minres	none	10000	0.668119	34.0389
	minres	gamg	14	0.140452	0.003246		minres	gamg	7	0.119645	0.000921		minres	gamg	5	0.164452	0.001603
	fcg	none	122	0.025989	4.02E+06		fcg	none	2	0.000343	2.36E+07		fcg	none	2	0.159129	1.66E+10
	fcg	bjacobi	212	0.013543	7.84E+06		fcg	bjacobi	363	0.046787	7.81E+06		fcg	bjacobi	272	0.241707	1.58E+07
	chebyshev	none	4	0.015245	8.25E+07		chebyshev	none	3	0.003248	7.02E+08		chebyshev	none	3	0.058992	8.86E+07
	chebyshev	jacobi	39	0.122719	0.001879		chebyshev	jacobi	699	0.12092	0.004448		chebyshev	jacobi	1143	0.067775	nan
10X10	cg	none	2	0.00289	22910.3	40X40	cg	none	2	0.004312	937607	100X100	cg	none	2	0.006798	9.70E+06
	cg	bjacobi	2	0.021117	293.628		cg	bjacobi	2	0.020533	675.887		cg	bjacobi	2	0.356989	1136.8
	cg	asm	2	0.003584	295.083		cg	asm	2	0.035562	688.392		cg	asm	2	0.035876	1148.31
	cg	gamg	2	0.010605	930.332		cg	gamg	2	0.055821	4183.85		cg	gamg	2	0.095208	9039.14
	gmres	none	90	0.003702	0.003116		gmres	none	10000	0.449353	631.042		gmres	none	10000	0.770822	999.972
	gmres	bjacobi	19	0.011577	0.002142		gmres	bjacobi	68	0.051436	0.006675		gmres	bjacobi	191	0.319747	0.011036
	gmres	asm	14	0.017305	0.001429		gmres	asm	44	0.023278	0.006679		gmres	asm	111	0.069345	0.010971
	gmres	gamg	6	0.140962	0.008995		gmres	gamg	7	0.139114	0.010287		gmres	gamg	7	0.636205	0.030508
	minres	none	10000	0.372786	2.30442		minres	none	10000	0.375323	22.0898		minres	none	10000	0.508135	38.4623
	minres	gamg	9	0.010661	0.000746		minres	gamg	5	0.072168	0.008607		minres	gamg	5	0.195952	0.00084
	fcg	none	32	0.002098	1.95E+07		fcg	none	2	0.041966	6.60E+08		fcg	none	2	0.013272	4.61E+10
	fcg	bjacobi	182	0.1287	4.06E+06		fcg	bjacobi	362	0.223342	7.83E+06		fcg	bjacobi	370	0.205465	1.66E+07
	chebyshev	none	3	0.023563	2.44E+07		chebyshev	none	3	0.00117	1.94E+09		chebyshev	none	3	0.128153	8.26E+07
	chebyshev	jacobi	183	0.012918	0.003121		chebyshev	jacobi	1143	0.027686	nan		chebyshev	jacobi	1143	0.192794	nan

Performance Analysis

Nx	Solver	Preconditi	Iteration	Time(s)	Residuals
250X250	cg	none	1	0.13587	9.76E+07
	cg	bjacobi	2	0.204686	1853.97
	cg	asm	2	1.321608	1861.24
	cg	gamg	2	0.17497	22862.6
	gmres	none	10000	4.74191	1581.14
	gmres	bjacobi	601	0.274256	0.018478
	gmres	asm	507	0.500552	0.018522
	gmres	gamg	10	0.540266	0.061974
	minres	none	10000	1.693963	61.917
	minres	gamg	3	0.174803	0.0819
	fcg	none	1	0.0772	9.76E+07
	fcg	bjacobi	392	0.363333	4.32E+07
	chebyshev	none	4	0.532776	3.18E+09
	chebyshev	jacobi	1143	0.166981	nan



Conclusion & Learning Outcomes

- FDM, FEM and FVM results closely match the exact solution across all grid sizes.
- Residuals remains very low ($< 10^{-12}$) which shows good solver convergence
- Minor residual rise at finer grids is due to accumulated round-off errors.
- GMRES: Most accurate and robust solver; converged consistently across all grid sizes with ASM or GAMG preconditioners .
- CG: Fast for coarse grid but unstable and inaccurate for finer meshes
- MINRES: shows false convergence-low residuals but physically incorrect solution
- Chebyshev: Stable only for small grids; diverged for higher resolution
- FCG: Poor convergence and high residuals across all cases; not suitable for this problem.

Questions! & THANK YOU